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Statement of Purpose

of Samrat Kumar Dey (PhD in Informatics applicant for Fall 2025)

Since the beginning of my research journey in machine learning for healthcare, I have cultivated a deep passion for using advanced computing techniques to solve complex healthcare problems. My research interests, which lie at the intersection of artificial intelligence (AI) in healthcare, data-driven decision-making, electronic health record (EHR) analysis, and health data visualization, have evolved over the years, driven by my academic training and hands-on research experience. My goal in pursuing a PhD in Informatics at the University of Missouri, specifically in the specialization of health/clinical informatics, AI and computational methods, and data science, is to develop and implement cutting-edge data-driven AI and machine learning models to predict, detect, and diagnose emerging health issues, thereby improving patient outcomes and healthcare efficiency.

Research Background Over the last decade, I have had the privilege of engaging in various collaborative research projects globally that have broadened my expertise and honed my skills in AI, health data analytics/visualizations, machine learning, and healthcare applications. My journey toward artificial intelligence for health and healthcare informatics started during the COVID-19 pandemic under the guidance of my graduate advisor. I was more interested in exploring data-driven decision-making approaches with machine learning algorithms in low-resource settings, especially in countries like Bangladesh. My research focuses on predictive analytics for disease outbreaks and visual exploratory data analysis [1] of the COVID-19 pandemic, resulting in top journal publications, such as in the Journal of Medical Virology. Most notably, I formed and led a multidisciplinary research group during that period that became the first from the country to publish multiple contributions, including **understanding the dynamics of worldwide vaccination progress** [2], **R-CNN-based covid case detection from x-ray images** [3], **VGGFusionNet model to detect covid patients on CT images** [4], and **MobileNet-based face mask detection model** [5] in the form of prestigious journals and book chapters. I have also worked on understanding the epidemic **dynamics of COVID-19 in Southeast Asia**[6] based on the SIR model and also explored the **epidemiological nature of Bangladesh** [7] during the COVID-19 outbreak utilizing a data visualization approach.

Previous Research During my graduate study as a Research Assistant (RA) under my graduate supervisor Prof. Md. Mahbubur Rahman at **MIST**, I focused on Software Defined Networking (SDN) and developed an intrusion detection model [8] using deep learning algorithms. I applied feature selection techniques such as Info Gain, Gain Ratio, and the Chi-square test to analyze the NSL-KDD dataset. Later, I proposed a flow-based anomaly detection model using Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM) networks, further enhancing the model's performance for detecting network intrusions in SDN environments.

In the final year of my undergraduate studies at **PSTU**, I began contributing to research under the guidance of Prof. Golam Md. Muradul Bashir, culminating in a thesis focused on measuring the complexity of programming problems based on cognitive weight. I proposed a novel complexity model [9] designed to assist facilitators in distributing equivalent programming problems among learners. To validate and implement the model, I developed a software tool using Java Standard Edition (SE) and MySQL Database, which calculates cognitive complexity based on the proposed algorithm.

Current Research One of my most impactful projects has been on birth asphyxia detection through infant cry analysis utilizing the country's first infant cry dataset. In this project, funded by the Information and Communication Technology (ICT) Division of the Government of Bangladesh, I led a team in developing a machine-learning model initially using the Baby Chillanto dataset, with plans to integrate our developed dataset later. This project, which is still ongoing, aims to improve the early detection of neonatal asphyxia using deep learning approaches. We presented this work [10] at the AI for Good Global Summit, where it achieved the AI for Good-Innovate for Impact Award. I have also contributed to developing several machine learning models for healthcare data-driven decision-making, including systems for **dengue disease prediction** [11], **cervical cancer detection** [12], **diabetic retinopathy prediction** [13], and **chronic disease diagnosis** [14]. These experiences have sharpened my data preprocessing, feature extraction, and model optimization skills, which I plan to develop further in my PhD research.

Teaching & Outreach My preference for a career in academia, which promises a distinctive blend of teaching and mentorship opportunities and research, has influenced my decision to pursue a PhD. Since 2015, I have been deeply involved in academia as an **Assistant Professor** and researcher in the Computer Science and Engineering (CSE) departments of two different universities in Bangladesh. My current position is at the country's first open-distance education-based institution, one of the most prestigious universities with a long tradition of blended learning methodologies. The innovative approach of combining traditional and digital education models has enriched my teaching experience and inspired me to think creatively about how technology can transform education and healthcare. My teaching responsibilities, including courses in object-oriented programming, artificial intelligence, human-computer interaction, and machine learning, have allowed me to inspire the next generation of engineers while pursuing my research goals. Besides, I am serving as an Academic Editor for the PLoS ONE and Computational and Mathematical Methods journal, published by Wiley. At this moment, I am also serving as a Lead Guest Editor for a Special Issue titled Data-Driven Decision Making in Healthcare Informatics, which will be published by Wiley as part of the BioMed Research International Journal. Recently, I have been honoured with a nomination for the prestigious India Science and Research Fellowship 2024–25 by the **Bangladesh Academy of Science**, and **the Government of Bangladesh,** a highly selective recognition awarded to only eight individuals this year for their outstanding research contributions. I am also the recipient of a prestigious **scholarship** supported by the Bill and Melinda Gates Foundation, awarded as part of the *"Democratizing Public Health Modeling Using AI-based Tools"* grant to Child Health Research Foundation (CHRF).

Why MIZZOU Informatics? My primary intellectual interest is in the application of generative AI and large language models (LLMs) for disease predictive modeling and healthcare decision-making in the field of clinical/healthcare informatics. At the MU Institute for Data Science and Informatics under the supervision of **XXX**, I am particularly interested in how AI models can be leveraged to analyze EHR data to diagnose complex health conditions, especially in underserved populations with limited access to healthcare. The challenge of creating AI models that are both robust and interpretable is central to my work, as I believe these models can greatly improve healthcare delivery by providing clinicians with actionable insights while ensuring transparency and ethical use of data. In addition to disease diagnosis, I aim to explore data-driven clinical decision support systems for improving patient care in hospital settings by using large-scale EHR datasets and generative AI models. My vision is to create flexible systems that can assist healthcare professionals in making better clinical decisions by integrating patient history, current symptoms, and predictive models into a unified platform.

Besides, the PhD program in Informatics at the University of Missouri offers the ideal environment to achieve my research aspirations at the intersection of artificial intelligence, data science, healthcare informatics, and health data visualization. With its focus on cutting-edge research and its interdisciplinary blend of data science, geoinformatics, bioinformatics, and health informatics, the program aligns seamlessly with my academic background and professional goals. With its comprehensive research ecosystem and commitment to impactful innovation, the University of Missouri is the perfect place for me to grow as a scholar and create technologies that improve lives.

Career Goals My long-term goal is to contribute to academia and industry by advancing the field of AI and data-driven healthcare systems. Upon completing my PhD, I plan to pursue a career in research and development within academic institutions or healthcare technology companies, where I can continue to innovate and deploy AI models that address the most pressing healthcare challenges. I am particularly interested in developing scalable healthcare solutions for emerging economies, where resource constraints demand innovative disease diagnosis and treatment planning approaches. In summary, my strong academic foundation, extensive research experience, and passion for healthcare AI make me a suitable candidate for the PhD in Informatics program at the University of Missouri. I look forward to the opportunity to contribute to the program's vibrant research community and to work on impactful projects that will shape the future of healthcare technology.

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